Richard Whitehill

Curriculum Vitae

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Old Dominion University
Department of Physics
4402 Elkhorn Ave,
Norfolk, VA 23529

Thomas Jefferson National Accelerator Facility
Theory Center
12000 Jefferson Ave,
Newport News, VA 23606

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University E-mail: rwhit058@odu.edu
JLab E-mail: whiteric@jlab.org
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EDUCATION

Old Dominion University [ODU] 2023 –
- Ph.D. Student (GPA: 4.0/4.0)
- Research Assistant
- Supervision: Arkaitz Rodas-Bilbao, Nobuo Sato

Wichita State University [WSU] 2020 – 2023
- Bachelor's of Science (Summa Cum Laude, GPA: 3.985/4.0)
- Majors: Physics and Mathematics
- Minor: Computer Science
- Emory Lindquist Honors Scholar

Kansas Academy of Mathematics and Science [KAMS] 2018 – 2020
- Two-year dual enrollment program at Fort Hays State University [FHSU]

Eisenhower High School 2016 – 2020
- Summa Cum Laude (Rank: 1st of 208)

EMPLOYMENT

Thomas Jefferson National Accelerator Facility
- Undergraduate Student Research Intern 2021 – 2023
- Science Undergraduate Laboratory Internship [SULI] 2022

Old Dominion University 2021
- Research Experience for Undergraduates [REU]

FHSU PHYS 211L TA 2019 – 2020

AWARDS

- DOE Office of Science Graduate Student Research (SCGSR) Program Fellowship 2026
- JSA/JLab Graduate Fellowship 2025 – 2026
- Distinguished Performance on the Written Candidacy Exam 2024
- Brian Donald Diederich and Flavia Alexandra Osorhean Scholarship for Doctoral Research in Nuclear Physics 2023
- University Fellowship (Temple University) 2023
  - declined due to attending ODU
- University Graduate Fellowship (Penn State University) 2023
  - declined due to attending ODU

- Rosalee and Alvin Sarachek Award for Scholarly Excellence in the Natural Sciences 2023
- Dorothy and Bill Cohen Honors College Enhancement Scholarship 2023
- Wayne Pfeiffer Mathematics Endowed Scholarship 2022
- Douglas A. Knight Memorial Scholarship 2022
- Nelson S. Ladd Physics Scholarship 2021 – 2022
- Spirit Aerosystems National Merit Scholar 2020
- College Board National Hispanic Recognition Program Scholar 2020
- WSU Dean's List
  - Fall 2020, Spring 2021, Fall 2021, Spring 2022, Fall 2022, Spring 2023
- FHSU Dean's Honor Roll 2018 – 2020
  - Fall 2018, Spring 2019, Fall 2019, Spring 2020

SERVICE

- Hampton Roads Science Ambassador Program 2025
- Student Representative for KAMS Director Search Committee 2020

SEMINARS/CONFERENCES

- *On the reconstruction of SIDIS observables in the Breit frame* May 2026
  QCD Evolution 2026
- *Phenomenological estimate for the longitudinal structure function in SIDIS* May 2026
  DIS2026: 33$^{\text{rd}}$ International Workshop on Deep Inelastic Scattering and Related Subjects
- *Impact of PVDIS on the weak mixing angle and high-x parton distributions* May 2026
  DIS2026: 33$^{\text{rd}}$ International Workshop on Deep Inelastic Scattering and Related Subjects
- *Impact of parity-violating DIS on the nucleon strangeness and weak-mixing angle* May 2025
  QCD Evolution 2025
- *Exploring lepton-charge asymmetries and parity-violation in large-p_T SIDIS* March 2025
  Positron Working Group Workshop
- *Impact of parity-violating DIS on the nucleon strangeness and weak mixing angle* November 2024
  CTEQ Fall Meeting 2024
- *Impact of PVDIS with 22 GeV electrons at JLab on the nucleon strangeness* June 2024
  SoLID Opportunities and Challenges of Nuclear Physics at the Luminosity Frontier
- *Impact of parity-violating DIS on the weak mixing angle and nucleon strangeness* June 2024
  Hampton University Graduate Studies [HUGS] Program
- *Accessing gluon polarization in large-P_T SIDIS* March 2023
  DIS2023: XXX International Workshop on Deep-Inelastic Scattering and Related Subjects
- *Accessing Gluon Polarization in SIDIS* September 2022
  Computational and Applied Mathematics Seminar
- *Gluon Polarization in Large-P_T SIDIS* March 2022
  Wichita State University Physics Seminar

PUBLICATIONS

1. **R. M. Whitehill**, P. Anderson, L. Gamberg, W. Melnitchouk, N. Sato, and X. Zheng
   In preparation (2026)
   On the reconstruction of SIDIS observables in the Breit frame

2. A. Bacchetta, M. Cerutti, L. Gamberg, and **R. M. Whitehill**
   In preparation (2026)
   Phenomenological estimates of the unpolarized longitudinal structure function in SIDIS

3. T. C. Rogers and **R. M. Whitehill**
   In preparation (2026)
   What are the consequences of independent factorization and renormalization scales?

4. **R. M. Whitehill**, M. M. Dalton, T. Liu, W. Melnitchouk, and N. Sato
   Preprint (2026), arXiv:2606.26056 [hep-ph]
   Impact of parity-violating deep-inelastic scattering on the weak mixing angle and high-x parton distributions

5. **R. M. Whitehill**
   Submitted to Phys. Lett. B (2026), arXiv:2604.28154 [hep-ph]
   Mapping data sensitivities in global QCD analysis with linear response and influence functions

6. P. C. Barry, A. Prokudin, T. Anderson, C. Cocuzza, L. Gamberg, W. Melnitchouk, E. Moffat, D. Pitonyak, J.-W. Qiu, N. Sato, A. Vladimirov, and **R. M. Whitehill**
   Submitted to Phys. Rev. Lett. (2025), arXiv:2510.13771 [hep-ph]
   First simultaneous analysis of transverse momentum dependent and collinear parton distributions in the proton

7. J. Y. Araz, V. Mikuni, F. Ringer, N. Sato, F. Torales Acosta, and **R. M. Whitehill**
   Phys. Lett. B 868, 139694 (2025), DOI, arXiv:2410.22421 [hep-ph]
   Point cloud-based diffusion models for the Electron-Ion Collider

8. J. Karpie, **R. M. Whitehill**, W. Melnitchouk, C. Monahan, K. Orginos, J.-W. Qiu, D. G. Richards, N. Sato, and S. Zafeiropoulos
   Phys. Rev. D 109, 036031 (2024), DOI, arXiv:2310.18179 [hep-ph]
   Gluon helicity from global analysis of experimental data and lattice QCD Ioffe time distributions

9. **R. M. Whitehill**, Y. Zhou, N. Sato, and W. Melnitchouk
   Phys. Rev. D 107, 034033 (2023), DOI, arXiv:2210.12295 [hep-ph]
   Accessing gluon polarization with high-P_T hadrons in SIDIS